Family Tineidae

Clothes moths and relatives

Occurrence 2,500 spp. worldwide; in rotting wood, fungi, dried organic matter, woollens, fabrics, dry foodstuffs

Many of these small, drab moths are pests, and are variously known as grain moths, case-bearing moths, and tapestry moths. Typically dull brown in color, some have a shiny golden appearance. Their head has a covering of raised, hairlike scales or bristles and the proboscis is short or absent. The wings, $\frac{5}{16}$–¾ in (0.8–2 cm) across, are narrow and are held at a steep angle over the body at rest. Females lay 30–80 eggs over a period of 3 weeks. Caterpillars are saprophagous. Some of them make portable cases of silk and debris, others make a silk web wherever they feed. Adults do not generally feed. Many species will not readily fly, scuttling away from danger instead.

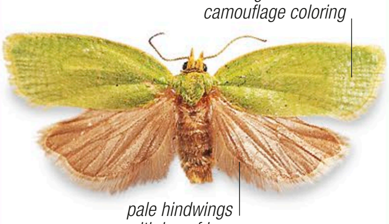
TINEOLA BISSELLIELLA
Also known as the common clothes moth, this widespread species inhabits clothing, carpets, and furs. The female lays eggs in the folds of clothes.

Family Tortricidae

Tortricid moths

Occurrence 10,000 spp. worldwide; on leaves, shoots, buds, fruit

These small moths are also known as bell moths. While most are brown, green, or gray, and patterned to blend in with bark, lichen, and leaves, some can be brightly colored. Their head is covered with rough scales. The wingspan is $\frac{5}{16}$–1½ in (0.8–3.4 cm) and the forewings, unusually rectangular, are held rooflike over the body at rest. A female can lay up to 400 eggs in a week, singly or in small groups, on the surface of fruit or leaves. The caterpillars feed on leaves, shoots, and buds of plants and often tie or roll leaves with silk. Being fruit- and stem-borers or gall-formers, many species are pests that cause damage to trees and fruit crops.



CLEPSIS RURINANA
Found in woodland and forests of Europe and Asia, the caterpillars of this species roll the leaves of deciduous trees and feed inside.

Family Uraniidae

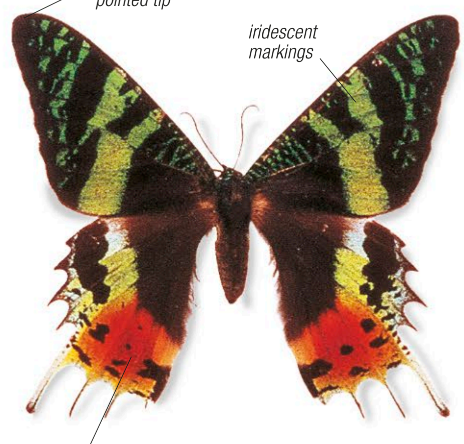
Uraniid moths

Occurrence 700 spp. in tropical and subtropical regions; on plants, especially euphorbias

This family includes large, long-tailed, colorful, day-flying moths with iridescent wing scales, and dull, nocturnal species without tails. The South American and Malagasy species are so colorful and large that they are often mistaken for butterflies. The wings typically span 2¼–4 in (6–10 cm). Adults are often migratory, sometimes forming large swarms, in response to changes in larval food quality. The caterpillars of many species eat plants of the family Euphorbiaceae. They avoid ant predation by dropping from leaves on silk lines.

ALCIDES METAURUS
This day-flying zodiac moth from the rain forests of North Queensland, Australia, feeds on flowers during the day and flies back to the canopy in the evening.

LYSSA ZAMPA
This large, Southeast Asian moth has a wingspan of 4–4¼ in (10–10.6 cm), a little larger than most species in the family.



CHRYSIRIDIA RHIPHEUS
Also known as the Madagascan sunset moth, this is a brightly colored day-flying species that inhabits woodland and forests of Madagascar.

Order Hymenoptera

Bees, wasps, ants, and sawflies

The order Hymenoptera is a vast group of insects that contains about 198,000 species in 91 families. It is divided into 2 suborders: plant-eating sawflies (Symphyta); and wasps, bees, and ants (Apocrita), which—unlike sawflies—have a narrow "waist" and, in females, an ovipositor that may sting. Most species have 2 pairs of membranous wings, joined in flight by tiny hooks. Ecologically, these insects are of tremendous importance in acting as predators, parasites, pollinators, or scavengers. Social Hymenoptera, including ants and bees, are the most advanced insects on earth.

Family Agaonidae

Fig wasps

Occurrence 750 spp. in subtropical and tropical regions; on fig trees

Fig wasps have evolved a symbiotic relationship with fig trees. The female wasps, no more than ⅛ in (3 mm) long, crawl inside figs, which are lined with tiny flowers. They pollinate some flowers, and lay eggs in others. After developing, the wingless males fertilize the females, which fly away to find new trees.

BLASTOPHAGA PSENES
The common fig wasp is found in Asia, southern Europe, Australia, and California. It pollinates Ficus carica*, the common fig. Females are 20 times more common than males.*

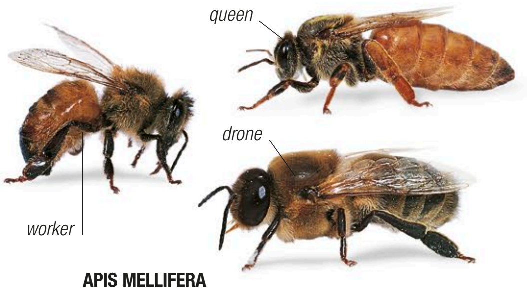
Family Apidae

Honey bees and relatives

Occurrence 5,700 spp. worldwide; in all well-vegetated, flower-rich areas

This family includes one of the world's most useful insects—the honey bee—together with orchid bees and bumble bees. They provide honey and wax, and pollinate most of the world's plants. Typically, they are ⅛–1 in (0.3–2.7 cm) long. Most females have a venomous sting and a basket to carry pollen on the hindlegs. Honey bees and bumble bees are social insects, living in complex colonies consisting of a queen, males, and sterile female workers. The honey bee nest is a vertical array of wax combs divided into thousands of cells for rearing the young and storing pollen and honey. Queens may lay more than 100 eggs a day. Bumble bee nests are made of grass with wax cells, and the queens raise fewer offspring.

BOMBUS TERRESTRIS
Like all its relatives, the European buff-tailed bumble bee has a body covered by "fur," enabling it to fly in the cool conditions of early spring.



APIS MELLIFERA
Originally from Southeast Asia, the honey bee is now raised all over the world. A honey bee colony has strict division of labor. The queen lays all the colony's eggs, while the workers gather food and tend the larvae. The queen is fertilized by a drone.